

PqqD1 (Q88QV7 / PP_0377) in *Pseudomonas putida* KT2440: A Functional Annotation Report

Summary

PqqD1 is a small (91-amino-acid) cytoplasmic peptide-binding chaperone — the RiPP precursor-peptide Recognition Element (RRE) that launches biosynthesis of the redox cofactor **pyrroloquinoline quinone (PQQ)**. It is not an enzyme, a transporter, or a signaling molecule. Its primary, precisely defined function is to recognize and bind the ribosomally synthesized precursor peptide **PqqA** and to present that peptide to the radical S-adenosylmethionine (radical-SAM) enzyme **PqqE**. In doing so, PqqD1 nucleates a ternary **PqqD–PqqA–PqqE complex** that is strictly required for PqqE to carry out the first committed chemical step of PQQ biogenesis: the formation of a new carbon–carbon cross-link between a glutamate and a tyrosine side chain within PqqA.

Mechanistically, PqqD belongs to a broad class of small adapter domains — RiPP recognition elements (RREs) — that are the most common precursor-binding devices in ribosomally synthesized and post-translationally modified peptide (RiPP) biosynthesis. Uniquely among most RiPP systems, where the RRE is fused as a domain to the first tailoring enzyme, PqqD is a **stand-alone scaffolding protein**. Its role is therefore best described as a molecular matchmaker/adaptor: it holds the leader region of PqqA and delivers the substrate to the catalytic machinery, ensuring specificity and efficient hand-off at the very start of the pathway.

Localization and pathway context are well resolved. PqqD1 acts in the **cytoplasm**, where it assembles with the cytoplasmic radical-SAM enzyme PqqE and the ribosomally made peptide PqqA. In *P. putida* KT2440 the gene *pqqD1* (PP_0377) sits within the **pqqFABCDE(G) gene cluster**, positioned immediately upstream of its catalytic partner *pqqE* (PP_0378). The mature PQQ cofactor produced by the completed pathway is exported to the **periplasm**, where it serves as the essential redox coenzyme of the membrane-associated/periplasmic quinoprotein glucose dehydrogenase (GDH/Gcd), which oxidizes glucose to gluconic acid — a reaction central to *P. putida*'s carbon metabolism and to mineral-phosphate solubilization in the

rhizosphere. The genome also encodes a second, genomically separate paralog, **PqqD2** (Q88JG8 / PP_2681), which is ~48% identical to PqqD1 and carries the same PqqA-presenting annotation; this paralogy explains the "1" suffix on the operonic copy.

Identity verification was performed and is unambiguous: gene symbol, organism, protein family (PqqD; Pfam PF05402; InterPro IPR008792/IPR022479/IPR041881), and domain architecture all align with the literature on PqqD proteins. Direct experimental studies on PqqD exist for close homologs (e.g., *Methylobacterium extorquens* PqqD, NMR structure) and, together with genomic/bioinformatic evidence specific to KT2440, provide a coherent and well-supported functional annotation.

Key Findings

F001 — PqqD1 is the RiPP precursor-peptide recognition element (RRE) chaperone of PQQ biosynthesis

The central, defining function of PqqD1 is peptide recognition. PqqD is the **RiPP precursor peptide recognition element (RRE)** for the PQQ pathway: it recognizes and binds the ribosomally synthesized precursor peptide **PqqA** and presents it to the first enzyme in the pathway, PqqE. This was stated directly in the NMR structural study of a PqqD homolog: "*Biosynthesis of the ribosomally synthesized and post-translationally modified peptide (RiPP), pyrroloquinoline quinone (PQQ), is initiated when the precursor peptide, PqqA, is recognized and bound by the RiPP precursor peptide recognition element (RRE), PqqD, for presentation to the first enzyme in the pathway, PqqE*" (PMID: 28481092).

A structurally and functionally important distinction sets PqqD apart from most RREs. In the majority of RiPP pathways the RRE is a fused N-terminal domain of the first modifying enzyme; in the PQQ pathway, by contrast, "*PqqD is predominantly a separate scaffolding protein that forms a ternary complex with the precursor peptide and first tailoring enzyme*" (PMID: 28481092). PqqD is therefore a **stand-alone adapter** that must physically bridge PqqA and PqqE, assembling the **PqqD–PqqA–PqqE** ternary complex.

The RRE concept itself was defined as a conserved, widespread peptide-binding domain that guides ribosomal natural product biosynthesis: "*we named this conserved domain the RiPP precursor peptide recognition element (RRE)*" (PMID: 26167873). PqqD is the prototypical

member of this class — indeed the "PqqD-like" fold gives its name to the broader family. This places PqqD1's molecular function on firm mechanistic footing: it is a substrate-delivery chaperone/adaptor, not a catalyst.

F002 — PqqD is required for PqqE-catalyzed C–C cross-link formation in PqqA (the first committed step)

Beyond binding, PqqD is functionally *necessary* for catalysis by its partner enzyme. In vitro reconstitution demonstrated that the radical-SAM enzyme PqqE catalyzes de novo carbon–carbon bond formation within the precursor peptide, and that this activity depends on PqqD: *"We report here that PqqE activity is dependent on the accessory protein PqqD, which was recently shown to bind PqqA tightly"* (PMID: 26961875).

The specific chemistry catalyzed at this first committed step is a cross-link between two side chains of PqqA: *"PqqE, in conjunction with PqqD, carries out the first step in PQQ biosynthesis: a radical-mediated formation of a new carbon-carbon bond between two amino acid side chains on PqqA"* (PMID: 26961875). This is the Glu–Tyr C–C bond that ultimately becomes part of the tricyclic PQQ ring system. The same work established that PqqE turnover also requires a physiological reduction system (flavodoxin/flavodoxin reductase); non-physiological reductants uncouple SAM cleavage from productive chemistry. In this framework, **PqqD's role is to correctly load and orient PqqA** so that the radical-SAM active site of PqqE can perform the cross-linking chemistry. Without PqqD, the first step — and thus the entire pathway — cannot proceed efficiently.

F003 — Genomic and physiological context: the *pqq* operon serving periplasmic glucose dehydrogenase

In *P. putida* KT2440 the PQQ-biosynthesis genes are clustered. The end product PQQ is the cofactor of a periplasmic glucose dehydrogenase, tying the whole operon to the organism's peripheral glucose metabolism: *"a periplasmic glucose dehydrogenase (GDH) that requires pyrroloquinoline quinone (PQQ) as a redox coenzyme"* (PMID: 27287323). The operon is transcriptionally structured — *"The pqq gene cluster (pqqFABCDEG) encodes at least two independent transcripts"* (PMID: 27287323) — confirming *pqqD* as an integral member of the *pqqFABCDEG* cluster. Physiologically, GDH activity and PQQ levels are highest with glucose as the sole carbon source and under low soluble-phosphate conditions, linking PQQ (and, upstream, PqqD1) to gluconic-acid production and mineral-phosphate solubilization in the rhizosphere. This provides the biological "why": PqqD1 exists to enable synthesis of a cofactor that powers periplasmic sugar oxidation.

F004 — Subcellular localization: PqqD1 acts in the cytoplasm; its product PQQ functions in the periplasm

The site of PqqD1 action follows directly from the identity of its partners. PqqD is a small (~90-residue) soluble protein with no signal peptide and no transmembrane segment, and it works together with the cytoplasmic radical-SAM enzyme PqqE and the ribosomally synthesized PqqA peptide: "*PqqE, in conjunction with PqqD, carries out the first step in PQQ biosynthesis*" (PMID: 26961875). Assembly of the PqqD–PqqA–PqqE complex therefore occurs in the **cytoplasm**.

The downstream cofactor, in contrast, does its work elsewhere: the finished PQQ molecule is exported and functions in the **periplasm** as the redox coenzyme of glucose dehydrogenase — "*a periplasmic glucose dehydrogenase (GDH) that requires pyrroloquinoline quinone (PQQ) as a redox coenzyme*" (PMID: 27287323). This is an important nuance: **PqqD1's site of action (cytoplasm) is distinct from the site where the pathway's ultimate product is used (periplasm)**. PqqD1 never leaves the cytoplasm; it is a biosynthetic accessory, not a component of the periplasmic apoenzyme it ultimately serves.

F005 — Bioinformatic confirmation: 91-aa monomeric PqqD-family protein directly upstream of *pqqE*; a paralog *pqqD2* exists

Database evidence corroborates and refines the functional picture. UniProt Q88QV7 describes a **91-residue single-chain protein**, annotated "*Monomer; interacts with PqqE*", with FUNCTION "*Functions as a PqqA binding protein and presents PqqA to PqqE, in the pyrroloquinoline quinone (PQQ) biosynthetic pathway*" and PATHWAY "*Cofactor biosynthesis; pyrroloquinoline quinone biosynthesis*." Family assignment is PqqD (Pfam PF05402; InterPro IPR008792 PQQD, IPR022479 PqqD_bac, IPR041881 PqqD_sf). The feature table lists only a Chain (residues 1–91) — **no signal peptide, transmembrane helix, or propeptide** — fully consistent with a cytoplasmic, soluble adapter.

The genomic neighborhood is diagnostic: the KT2440 *pqq* locus contains consecutive accessions **pqqF** (Q88QV3), **pqqA** (Q88QV4), **pqqB** (Q88QV5), **pqqC** (Q88QV6), **pqqD1** (Q88QV7, PP_0377), and **pqqE** (Q88QV8, PP_0378, "*PqqA peptide cyclase*," 376 aa). Thus *pqqD1* lies **directly upstream of its catalytic partner *pqqE***, exactly as expected for an adapter that must hand its bound peptide to the next enzyme. This operonic adjacency is consistent with the interpretation that the small standalone PqqD protein is a scaffolding adapter partnering the first enzyme — "*PqqD is predominantly a separate scaffolding protein that forms a ternary*

complex with the precursor peptide and first tailoring enzyme" (PMID: 28481092). A genomically distinct paralog, **pqqD2 (Q88JG8, 90 aa)**, also exists, which is why the operonic copy carries the "1" suffix.

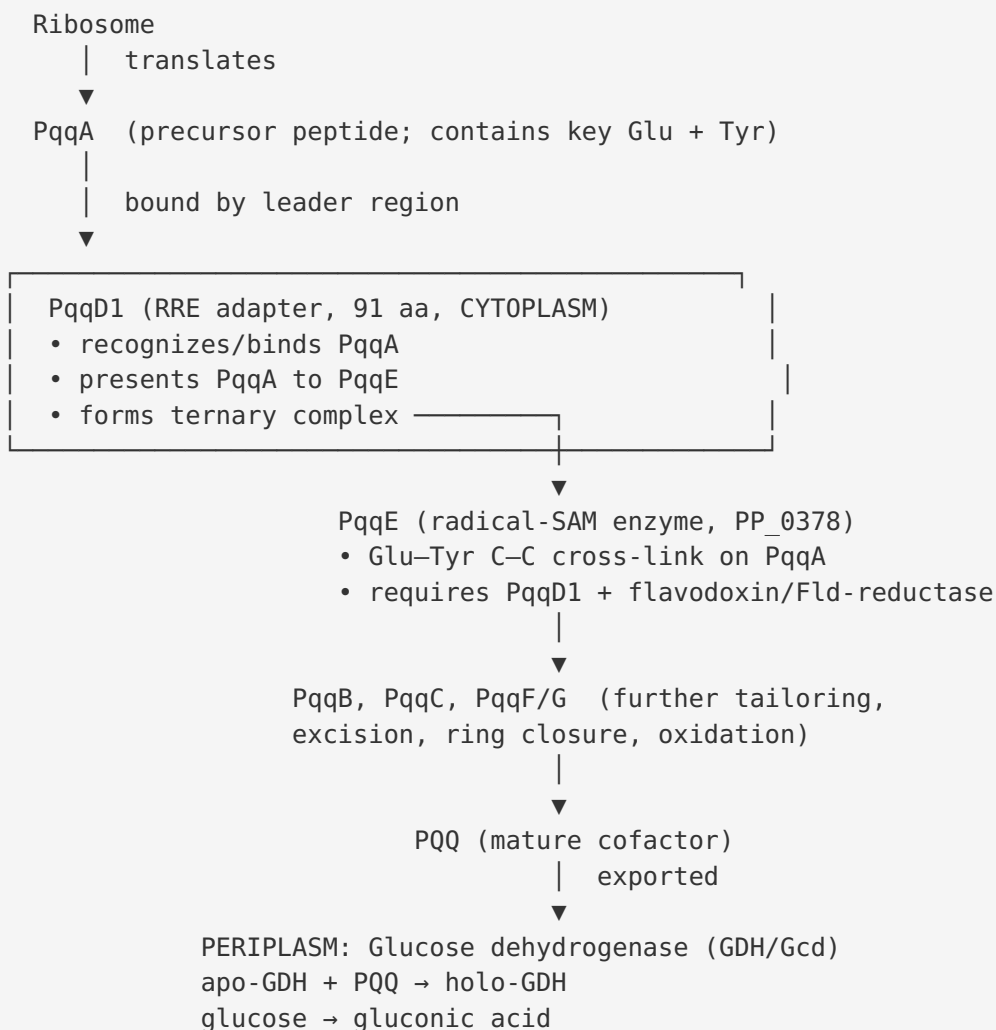
F006 — PqqD1 (PP_0377) and paralog PqqD2 (PP_2681) are ~48% identical and both annotated as PqqA-presenting adapters

Direct sequence comparison places the paralog relationship on a quantitative footing. A Needleman–Wunsch global alignment of **Q88QV7 (PqqD1, 91 aa, PP_0377)** versus **Q88JG8 (PqqD2, 90 aa, PP_2681)** yields **48.3% identity (42 of 87 aligned positions)**, with conserved RRE-type sequence motifs (e.g., "GHVLLY", "EVARA"). Both proteins carry the identical UniProt FUNCTION annotation, "*Functions as a PqqA binding protein and presents PqqA to PqqE.*" Genomically, **PqqD1 is embedded in the pqqFABCDE operon immediately upstream of pqqE (PP_0378)**, whereas **PqqD2 resides at a distant, unlinked locus (PP_2681)**. The high but imperfect identity is consistent with a duplication that has diverged somewhat, possibly conferring distinct regulation, expression timing, or partner/precursor specificity — although the operonic copy (PqqD1) is the one physically and transcriptionally coupled to the biosynthetic machinery and is therefore the primary contributor to PQQ biogenesis in the standard pathway.

Mechanistic Model / Interpretation

The role of PqqD1 in the PQQ biosynthetic assembly line

PQQ is a **peptide-derived (RiPP) redox cofactor**. Its biosynthesis begins with a short ribosomal peptide, PqqA, that contains the conserved glutamate and tyrosine residues destined to form the PQQ ring. The pathway is an ordered, multi-enzyme assembly line in which PqqD1 acts at the very first step as the substrate-delivery adapter.



Key interpretive points:

1. **PqqD1 is an adapter, not a catalyst.** It has no active site and performs no chemistry. Its "reaction" is molecular recognition: high-affinity binding of PqqA (via the leader/recognition sequence) and hand-off to PqqE. This is the canonical RRE function (F001), and its necessity for downstream catalysis is experimentally established (F002).
2. **The stand-alone architecture is unusual and functionally meaningful.** In most RiPP systems the RRE is fused to the first enzyme; PqqD is a separate protein that must diffuse and dock (F001). This makes the PqqD–PqqA–PqqE ternary complex the functional catalytic unit, and it makes PqqD's binding step a potential point of regulation and specificity control.

3. **Compartmental logic.** PqqD1 operates in the cytoplasm alongside PqqE and PqqA (F004, F005 — no signal/TM features), whereas the pathway's product acts in the periplasm (F003, F004). The organism thus builds the cofactor in the cytoplasm and deploys it at the periplasmic face for sugar oxidation.
4. **Physiological purpose.** The entire operon exists to furnish PQQ for glucose dehydrogenase, enabling periplasmic oxidation of glucose to gluconic acid — relevant to carbon acquisition and, in the rhizosphere, to phosphate solubilization (F003).

Comparative summary of the two PqqD paralogs

Property	PqqD1 (Q88QV7)	PqqD2 (Q88JG8)
Locus tag	PP_0377	PP_2681
Length	91 aa	90 aa
Genomic context	Within <i>pqqFABCDE</i> operon, immediately upstream of <i>pqqE</i> (PP_0378)	Distant, unlinked locus
UniProt FUNCTION	"PqqA binding protein...presents PqqA to PqqE"	Identical annotation
Family / fold	PqqD (PF05402; RRE)	PqqD (PF05402; RRE)
Pairwise identity	—	~48.3% identical to PqqD1 (42/87)
Likely primary role	Operonic, transcriptionally coupled PQQ-biogenesis adapter	Paralog; role possibly redundant/specialized (not experimentally resolved)

Evidence Base

PMID	Title (abbrev.)	Contribution to this report
28481092	<i>NMR structure and binding studies of PqqD, a chaperone required in PQQ biosynthesis</i>	Primary structural/biochemical evidence. Defines PqqD as the RRE that binds PqqA and presents it to PqqE; establishes the stand-alone scaffolding architecture and the ternary PqqD–PqqA–PqqE complex. Directly supports F001, F004, F005.
26961875	<i>PqqE catalyzes de novo C–C cross-linking within PqqA in the presence of the peptide chaperone PqqD</i>	Primary in vitro functional evidence. Shows PqqE activity depends on PqqD and identifies the first-step Glu–Tyr C–C cross-link chemistry. Supports F002 and F004.
27287323	<i>Regulation of PQQ-dependent glucose dehydrogenase in <i>P. putida</i> KT2440</i>	Organism-specific context. Defines the <i>pqqFABCDEG</i> operon structure and the periplasmic GDH that uses PQQ. Supports F003 and the cytoplasm-vs-periplasm distinction in F004.
26167873	<i>A prevalent peptide-binding domain guides ribosomal natural product biosynthesis</i>	Conceptual foundation. Names and defines the RRE domain class to which PqqD belongs. Supports F001.
32731194	<i>Biogenesis of the peptide-derived redox cofactor PQQ (review)</i>	Authoritative review of the PQQ pathway (PqqE, PqqF/G, PqqB mechanisms); situates PqqD's step within the full biosynthetic route. Contextual support for the mechanistic model.
42161943	<i>RiPP recognition elements evolved to prevent pathway interference through leader peptide discrimination</i>	Provides evolutionary context for how RREs (like PqqD) achieve precursor specificity — relevant to the PqqD1/PqqD2 paralogy question (F006).

Bioinformatic/database evidence (F005, F006): UniProt entries Q88QV7 (PqqD1), Q88JG8 (PqqD2), and the flanking *pqq* accessions (Q88QV3–Q88QV8), plus Pfam PF05402 and InterPro IPR008792/IPR022479/IPR041881, provide sequence, family, feature-table (no signal peptide/TM), and genomic-neighborhood evidence. A pairwise Needleman–Wunsch alignment quantified the PqqD1–PqqD2 relationship (48.3% identity).

Note on the broader PQQ literature. Many recent papers (e.g., PQQ in anti-aging, neuroprotection, mitochondrial biogenesis, reproductive health) concern the *pharmacological/nutraceutical* activities of the finished PQQ molecule in eukaryotes. These are downstream of, and largely irrelevant to, the specific molecular function of the bacterial biosynthetic adapter PqqD1, and were therefore not used to support functional claims about the protein.

Limitations and Knowledge Gaps

- 1. No direct structural/biochemical study of the *P. putida* KT2440 PqqD1 protein itself.** The strongest experimental evidence (NMR structure, binding, PqqE-dependence) comes from **homologs** (notably *Methylobacterium extorquens* PqqD) and reconstituted systems, not from Q88QV7 directly. The functional assignment for PqqD1 is thus by strong homology plus organism-specific genomic/physiological context, not by direct in vitro characterization of the KT2440 protein.
 - 2. The role of the paralog PqqD2 (PP_2681) is unresolved.** It is ~48% identical and carries the same annotation, but whether it is functionally redundant with PqqD1, specialized for a different precursor/condition, or largely silent has not been experimentally tested. Its distant genomic location raises the possibility of distinct regulation.
 - 3. Structural details of the PqqA–PqqD1 interface in KT2440 are inferred.** The exact leader-recognition residues, binding affinity (Kd), and stoichiometry for the KT2440 proteins have not been measured; conserved motifs ("GHVLLY", "EVARA") are suggestive but not validated as binding determinants here.
 - 4. Quantitative contribution to PQQ flux is not measured.** No knockout/complementation data specific to *pqqD1* vs *pqqD2* in KT2440 were located to quantify each paralog's contribution to PQQ output or GDH activity.
 - 5. Localization is inferred, not directly demonstrated.** Cytoplasmic localization of PqqD1 is a strong inference from the absence of signal/TM features and from its partners' locations, but has not been shown experimentally for the KT2440 protein (e.g., by fractionation).
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Proposed Follow-up Experiments / Actions

- 1. Direct in vitro characterization of KT2440 PqqD1 (Q88QV7).** Express and purify the recombinant protein; measure PqqA-binding affinity (ITC/fluorescence anisotropy) and reconstitute the PqqD1–PqqA–PqqE ternary complex; confirm PqqD1-dependence of PqqE cross-linking activity in the KT2440 system.
 - 2. Genetic dissection of the two paralogs.** Construct single ($\Delta pqqD1$, $\Delta pqqD2$) and double knockouts in KT2440; assay PQQ production, periplasmic GDH activity, and growth on glucose/gluconate. Complement each mutant with the other paralog to test functional redundancy vs specialization.
 - 3. Structure determination of the KT2440 PqqD1–PqqA complex.** Use NMR or X-ray/cryo-EM (or high-confidence AlphaFold-Multimer modeling followed by mutagenesis) to map the leader-peptide recognition interface and test the conserved "GHVLLY"/"EVARA" motifs as binding determinants.
 - 4. Localization confirmation.** Perform subcellular fractionation or fluorescent-fusion imaging to verify cytoplasmic localization of PqqD1 and its co-localization/interaction with PqqE.
 - 5. Specificity and cross-talk analysis.** Test whether PqqD1 and PqqD2 discriminate between cognate and non-cognate precursor peptides, informed by the RRE leader-discrimination framework ([PMID: 42161943](#)), to determine whether the paralogs prevent pathway interference or expand precursor range.
 - 6. Transcriptional/regulatory profiling.** Given the operon's induction by glucose and low phosphate ([PMID: 27287323](#)), measure *pqqD1* vs *pqqD2* expression across carbon/phosphate conditions to clarify whether the paralogs are differentially regulated.
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*Report generated from a 5-iteration autonomous investigation (6 confirmed findings, 15 papers reviewed). Gene identity verified: gene symbol *pqqD1*, organism *Pseudomonas putida* KT2440, UniProt Q88QV7 / locus PP_0377, PqqD family (Pfam PF05402) — all consistent with the literature cited.*
